Model Validation, Overfitting Control & Hyperparameter Tuning

1. Introduction

In this project, we implemented a machine learning pipeline for house price prediction using the California Housing dataset. The main objective was to improve model reliability by applying cross-validation, detecting overfitting, and tuning hyperparameters.

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2. Overfitting Analysis

Overfitting occurs when a model performs very well on training data but poorly on unseen test data. In this project, we used a Decision Tree model to demonstrate overfitting.

We observed that:

- Training RMSE was very low
- Test RMSE was significantly higher

This indicates that the model memorized training data instead of learning general patterns.

To reduce overfitting, we:

- Used cross-validation
- Applied hyperparameter tuning (max_depth, min_samples_split)

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3. Cross-Validation

Instead of relying on a single train-test split, we used 5-fold cross-validation.

Benefits:

- Provides more reliable performance estimate
- Reduces dependency on one data split
- Helps in selecting better models

The cross-validation RMSE gave a stable and realistic measure of model performance.

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4. Hyperparameter Tuning

We used GridSearchCV to find the best parameters for the Decision Tree model.

Parameters tuned:

- max_depth
- min_samples_split

This helped:

- Control model complexity
- Reduce overfitting
- Improve generalization

The best parameters were selected based on cross-validation performance.

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5. Model Comparison

We compared three models:

- Linear Regression
- Ridge Regression
- Tuned Decision Tree

Observations:

- Linear Regression is simple but less flexible
- Ridge Regression improves stability using regularization
- Tuned Decision Tree achieved the best performance after controlling overfitting

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6. Final Model Selection

The Tuned Decision Tree model was selected because:

- It achieved the lowest RMSE
- It had better R^2 score
- Overfitting was reduced after tuning
- Cross-validation results confirmed its reliability

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7. Conclusion

This project demonstrated the importance of model validation and tuning in machine learning. Instead of focusing only on accuracy, we improved model reliability and generalization using cross-validation and hyperparameter tuning.

This approach reflects real-world industry practices in building robust machine learning models.